

Representing Inequalities on a Number Line

AQA · KS4 Year 9–10

Answer each question in your workbook. Draw each number line carefully.

VISUAL EXAMPLE



$x < 5$ on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

Filled circle for $<$ or $>$ (should be open).

Open circle for $\leq$ or $\geq$ (should be filled).

Arrow pointing the wrong way.

$-2 \leq x$ is the same as $x \geq -2$.

METHOD

- 1 Find the boundary number
- 2 Open for $</>$, filled for $\leq/\geq$
- 3 Arrow towards the true values
- 4 Shade between for 'between' inequalities
- 5 Read back to check

WORKED EXAMPLE

Represent $x \geq -2$.

$\geq$ includes -2, so a filled circle; arrow points right.

Read-back: filled at -2, arrow right = $x \geq -2$

VISUAL AID

Open circle means the boundary value is NOT included ($<$ or $>$). Filled circle means it IS included ($\leq$ or $\geq$). The arrow points towards all the numbers that make the inequality true.

1. (2 marks)

Show on a number line: a) $x > 1$ b) $x \leq 4$

2. (2 marks)

Show on a number line: a) $x \geq -3$ b) $x < 0$

3. (2 marks)

Write the inequality: a) open circle at 2, arrow right b) filled circle at -1, arrow left

4. (3 marks)

Show on a number line: a) $x \leq 5$ b) $x < -4$ c) $-2 \leq x$

5. (3 marks)

Solve, then show on a number line: a) $2x + 1 > 7$ b) $3x - 2 \leq 10$ c) $5 - 2x \geq 1$

6. (3 marks)

Write and show: a) the number n is at least -1 b) the temperature t ($^{\circ}\text{C}$) is below 5

7. (3 marks)

Filled circle at -3, open circle at 4, shaded between. a) Write the inequality. b) List the integers.

8. (5 marks)

Machine safe when $t \geq 10$ and $t < 35$. a) Show each on a number line. b) Combine into one inequality. c) Is 35°C safe? Explain.